ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

LAWRENCE

MARKING SCHEME

JUNE 2013

CHEMISTRY 9189/2

Reversible rxn.

1 (a) ~~Equilibrium~~ with the rate of forward reaction equal to the rate of reverse reaction; / AW. [1]

(b) (i) pressure falls, equilibrium shifts to the left / AW; [1]

(ii) equilibrium position shifts to the right; / AW. [1]

(c)

- measure absorbance of rxn mixture at ~~different~~ fixed time intervals; [1]
- plot a graph of absorbance against time; (A) a sketch diagram with details of [1]
- the gradient at any time of the graph gives the rate of reaction; [1]

(d) (i) no effect; reaction has no volume change;

(A) Particles move together more effective collisions

(ii) fraction of molecules with $E > E_a$ increases; [1]

HI adsorbs on the catalyst / lowering of E_a

collision density increases; increasing rate of decomposition; of HI [1]

(iii) fraction of molecules with $E > E_a$ decreases; [1]

reduced collision frequency; rate of decomposition decreases; / less KE [1]

2 (a) (i) The potential difference of a half cell measured relative to a standard hydrogen electrode under standard conditions;A/W 298K, 1Molar, 1atm. [1]

(ii) - Inert metal connection / does not react with the electrolyte; [1]

- Increased surface area for equilibrium to be established; [1]

(b)

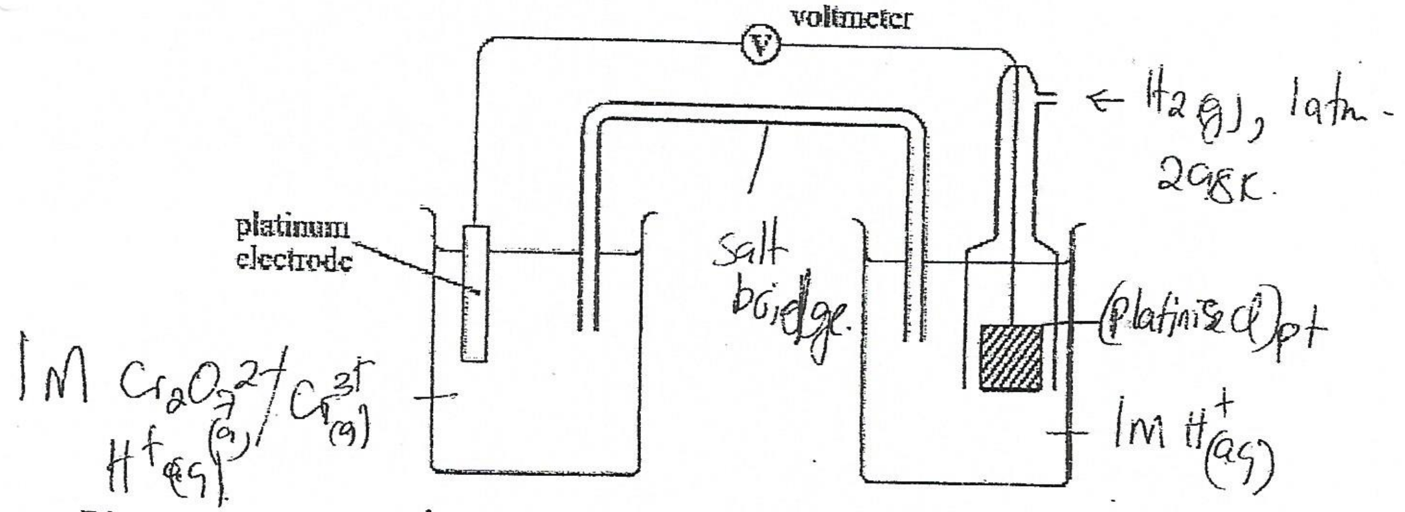


Diagram [1]

Labels [2]

8 labels. (2)

4-7 (1).

(c) E° values valid only for solutions of unit concentrations and measurements taken at atmosphere and 25°C. / A W. [1]

3 (a) (i) E(F – F) = 158 $kJmol^{-1}$

E(Cl – Cl) = 244 $kJmol^{-1}$

E(H – F) = 562 $kJmol^{-1}$

E(H – Cl) = 431 $kJmol^{-1}$

all of them. Quoted values of bond energies [1]

Fluorine has lower bond dissociation energy than chlorine but forms Stronger H – F bonds, hence reaction; is more exothermic [1]

(ii) E(I –I) = = 151 $kJmol^{-1}$

E(H – I) = 299 $kJmol^{-1}$

(1) mark. ; Qoted values of bond energy

H – I bond weak – require less energy [1].

~~Decrease in bond energy of H – I;~~ reaction less exothermic

Iodine reacts reversibly at 450°C in the presence of a catalyst; [1]

(b) (i) decreases down the group / A W. [1]

(ii) decrease in electronegativity / more electronegative more reactive; [1]

Ⓐ E° value becoming less positive.

(c) volatility decreases A ; [2]

strength of van der waal forces increases from F_2 to I_2 due to increase in number of constituent electrons. or molecular size. Ⓡ Molecular weight, molecular mass. [1]

(d) (i) 1 – loss of food [1]

2 No recreation due to excessive plant growth.

(ii) Avoid overuse of N fertilizers. [2]

use of manure.

Avoid sewage polluting water bodies.

Avoid stream bank cultivation;

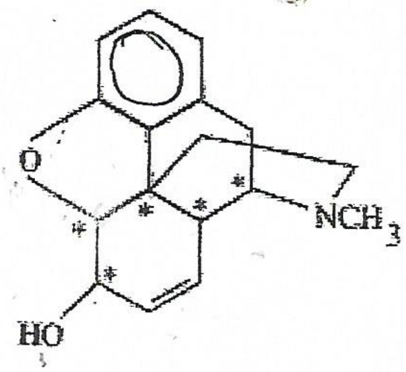
(e) Very strong N≡N bond / High bond energy [2]

[1]

• Nitrates cause oxidation of iron in blood.

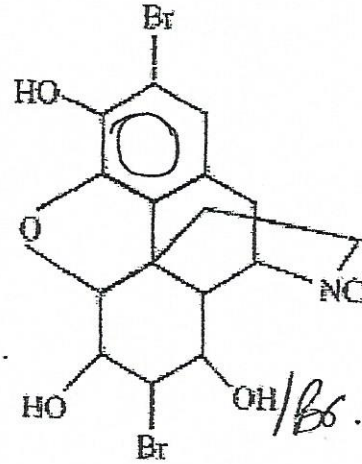
4 (a) (i) phenol, alkene/carbon to carbon double bond.
hydroxyl, amine;
(ether)
1 mark for any two functional groups [2]

(ii)



1 mark for all chiral centres shown; [1]

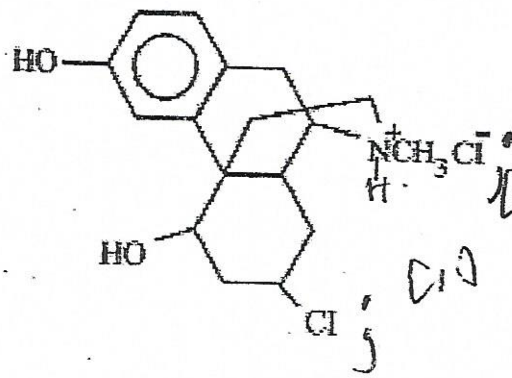
(b) (i)



/Br.

~~Addition of Br on aromatic ring;~~ [1]

(ii)

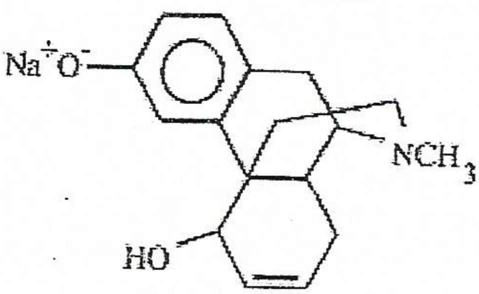


[1]

[1]

(~~Addition of Br and OH on the double bond;~~) [1]

(iii)



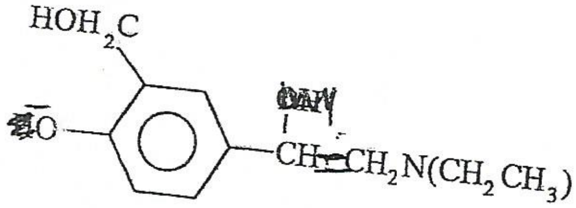
(c) neutralisation / acid base reaction [1]

5 (a) (i) aqueous bromine decolourised;
a white precipitate formed; [1]

(ii) colour changes from orange to green; [1]

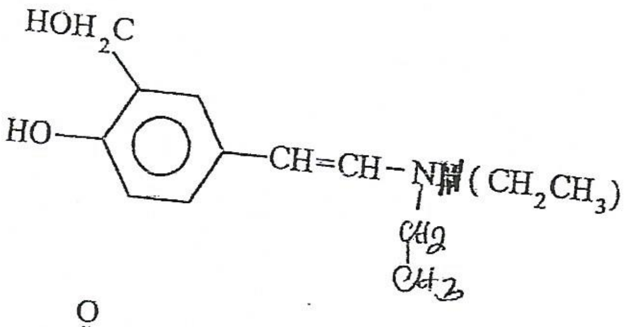
(iii) white fumes; [1]

(b) (i)



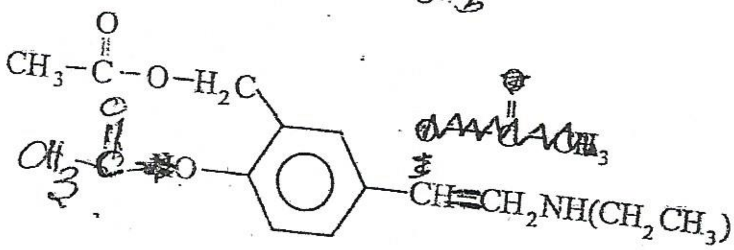
[1]

(ii)



[1]

(iii)



[1]

(c)
- food flavourants;
- perfumes;
- adhesives;
- solvents; any two

[2]